Mohammad Taha **Askari**

PhD CANDIDATE · ELECTRICAL AND COMPUTER ENGINEERING, THE UNIVERSITY OF BRITISH COLUMBIA

Electrical and Computer Engineering Department, The University of British Columbia, Vancouver BC V6T 1Z4, Canada

mohammadtaha@ece.ubc.ca | Homepage | Github | Google Scholar | Linkedin

Summary

PhD candidate in Electrical and Computer Engineering at the University of British Columbia, with a research focus on machine learning, probabilistic modeling, and sequence models. My work centers on designing and training neural sequence models to learn joint probability distributions in systems with memory and uncertainty. I have extensive experience conducting independent research, publishing in peer-reviewed venues, and building end-to-end ML prototypes in Python that integrate theory, experimentation, and scalable implementation.

Research Experience

M.A.Sc. & PhD Thesis at Data Communications Group
ECE Department, UBC

RESEARCH ASSISTANT (SUPERVISOR: PROF. LUTZ LAMPE) *Sep. 2020 - Apr. 2027*

- Conducted independent research on **probabilistic modeling and sequence design** for communication systems with memory and nonlinear dynamics.
- Developed an **analytical low-pass filter model** to characterize interactions between probabilistic shaping and nonlinear optical fiber channels, enabling interpretable system-level optimization.
- Proposed a **sequence selection algorithm** that improves robustness to nonlinear interference by exploiting temporal dependencies.
- Designed a **Bayesian carrier phase recovery algorithm** incorporating prior distributions, achieving improved performance in low-SNR regime.
- Investigated **end-to-end learning approaches** using sequence-based autoencoder architectures for probabilistic shaping and joint distribution learning.

Research Internship at Nokia Bell Labs
Greater Paris Metropolitan Region, France

MACHINE LEARNING RESEARCH INTERN *April. 2025 - April. 2026*

- Designed and implemented **Neural Probabilistic Shaping (NPS)**, an autoregressive **sequence modeling framework** using neural sequence models and **Gumbel-Softmax** sampling for joint distribution learning.
- Built an **end-to-end differentiable training pipeline** integrating neural models with differentiable channel models, enabling gradient-based optimization over sequences.
- Evaluated the proposed models on **multi-dimensional settings**, demonstrating improved robustness and scalability compared to heuristic and sampling-based methods.

Research Internship at Roche Canada
Greater Toronto Area, Canada

ALGORITHM R&D SOFTWARE ENGINEERING INTERN *Jun. 2024 - Oct. 2024*

- Designed and implemented **ML pipelines** for feature extraction, dataset generation, and neural network training for large-scale sequencing data.
- Developed and optimized **weakly supervised deep learning models**, including a custom **multi-label architecture** for multi-class anomaly detection and segmentation.
- Integrated **model interpretability, performance analysis, and computational complexity evaluation** into the training workflow to support iterative model improvement.

Publications and Posters

- M.T. Askari, L. Lampe, and A. Ghazisaeidi ”Neural Probabilistic Amplitude Shaping for Nonlinear Fiber Channels,” 2026 Optical Fiber Communications Conference and Exhibition (OFC).
- M.T. Askari, L. Lampe, and A. Ghazisaeidi ”Neural Probabilistic Shaping: Joint Distribution Learning for Optical Fiber Communications,” 2025 European Conference on Optical Communication (ECOC). [Online Access]
- M.T. Askari and L. Lampe, ”Probabilistic Shaping for Nonlinearity Tolerance,” Journal of Lightwave Technology. [Online Access]
- M.T. Askari and L. Lampe, ”Perturbation-based Sequence Selection for Probabilistic Amplitude Shaping,” 2024 European Conference on Optical Communication (ECOC). [Online Access]
- M.T. Askari and L. Lampe, ”Bayesian Phase Search for Probabilistic Amplitude Shaping,” 2023 European Conference on Optical Communication (ECOC). [Online Access]

- M.T. Askari, "Interplay between Fiber Nonlinearity and Probabilistic Amplitude Shaping," Master's Thesis, UBC [Online Access]
- M.T. Askari, L. Lampe, and J. Mitra, "Probabilistic Amplitude Shaping and Nonlinearity Tolerance: Analysis and Sequence Selection Method," Journal of Lightwave Technology. [Online Access]
- M.T. Askari, L. Lampe, and J. Mitra, "Nonlinearity Tolerant Shaping with Sequence Selection," 2022 European Conference on Optical Communication (ECOC) [Online Access]
- M.T. Askari, "Nonlinearity Tolerant Sequence Selection," Poster at 17th Canadian Workshop on Information Theory (CWIT)

Research Interests

Machine Learning for Sequential and Structured Data
Probabilistic Modeling and Joint Distribution Learning
Neural Sequence Models (Autoregressive Models, Transformers)
End-to-End Learning and Optimization
Information Theory and Statistical Inference

Education

PhD in Electrical & Computer Engineering — *Vancouver, Canada*
Electrical and Computer Engineering Department, The University of British Columbia (UBC) — *Sep. 2022 - May. 2027*
- Advisor: Prof. Lutz Lampe

M.A.Sc. in Electrical & Computer Engineering — *Vancouver, Canada*
Electrical and Computer Engineering Department, The University of British Columbia (UBC) — *Sep. 2020 - Aug. 2022*
- Cumulative GPA: 93.7 out of 100 (4.0/4.0) via 18 credits
- Advisor: Prof. Lutz Lampe

B.Sc. in Electrical Engineering — *Tehran, Iran*
Department of Electical Enginerring, Sharif University of Technology (SUT) — *Sep. 2015 - Jul. 2020*
- Cumulative GPA: 18.42 out of 20 (4.0/4.0) via 146 credits
- Advisor: Prof. Hamid Behroozi

Software Proficiency

Programming Python, C/C++, Git, SQL, Matlab, Julia, R

Relevant Courses

CPSC 436N-Natural Language Processing	*95.0 out of 100*	**Dr. Shwartz**	*UBC*	*Winter1 2023*
EECE 571D-Detection, Estimation, and Learning	*98.0 out of 100*	**Dr. Lampe**	*UBC*	*Winter2 2021*
EECE 562-Statistical Signal Processing	*90.0 out of 100*	**Dr. Wang**	*UBC*	*Winter1 2021*
CPSC 540-Advanced Machine Learning	*96.0 out of 100*	**Dr. Schmidt**	*UBC*	*Winter2 2020*
CPSC 340-Machine Learning & Data Mining	*100 out of 100*	**Dr. Wood**	*UBC*	*Winter1 2020*
EECE 565-Communication Networks	*95.0 out of 100*	**Dr. Wong**	*UBC*	*Winter1 2020*
Introduction to Machine Learning	*18.0 out of 20*	**Dr. Mohammadzade**	*SUT*	*Spring 2019*
Information & Coding Theory	*18.7 out of 20*	**Dr. Mirmohseni**	*SUT*	*Fall 2018*

Selected Course Projects

COVID-19 Detection: A Combination of Transformer Models with Image Augmentation Methods — *Winter 2020*
COURSE: Advanced Machine Learning — *UBC*
- Studied methods for generating synthetic X-ray images to improve the accuracy of COVID-19 detection models and implemented GAN and VAE to augment data.
- Applied transformer models to COVID-19 classification based on chest X-ray images.
- Supervisor: Dr. Schmidt